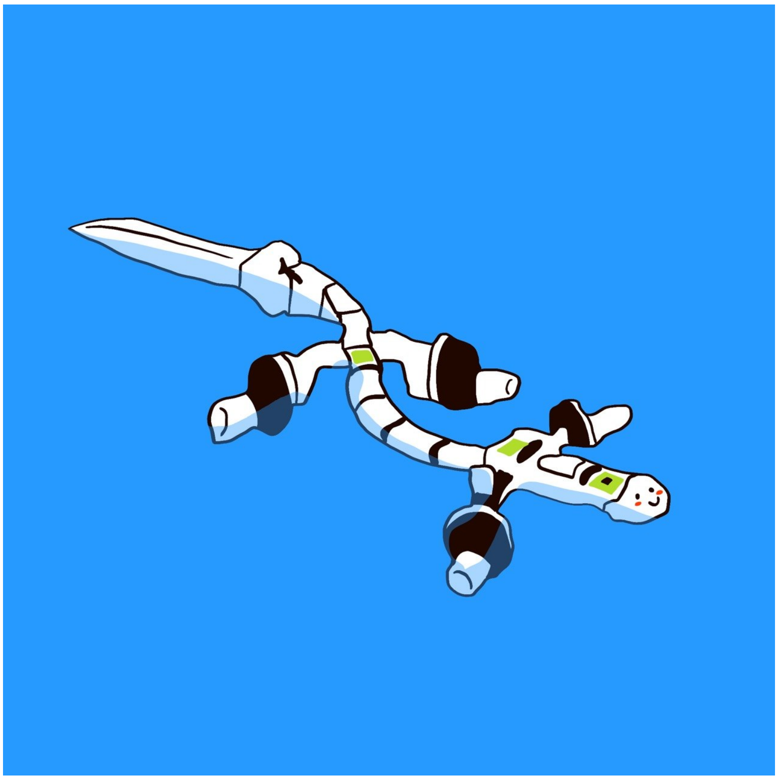
Amphibious Locomotion of Polymander

Computational Motor Control



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Project 1
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Introduction

In the context of the Computational Motor Control course, we designed a controller based on a Central Pattern Generator (CPG) and implemented sensory feedback in order to enable a salamander-like robot to achieve aquatic locomotion. This project allows us to study a model and the simulation of a robot performing swimming movements. Throughout this project, we will explore how CPGs and sensory feedback allow the robot to move effectively in an aquatic environment.

1 Wave Controller and Metrics

Question 1.2 : Implemention of the metrics (NM2, MM2, MM4)

After implementing the wave controller, we evaluate its performance using a set of metrics that characterize its behavior during swimming.

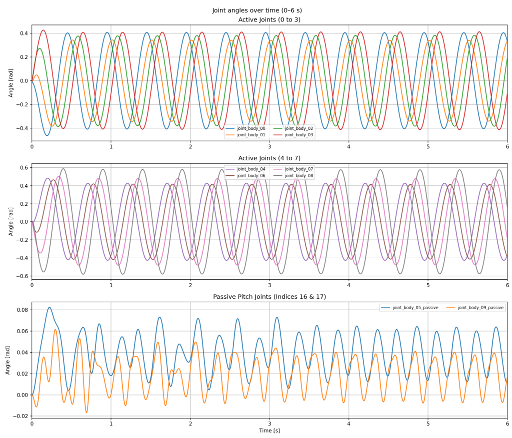


Figure 1: Evolution of joints angles for 0.6s

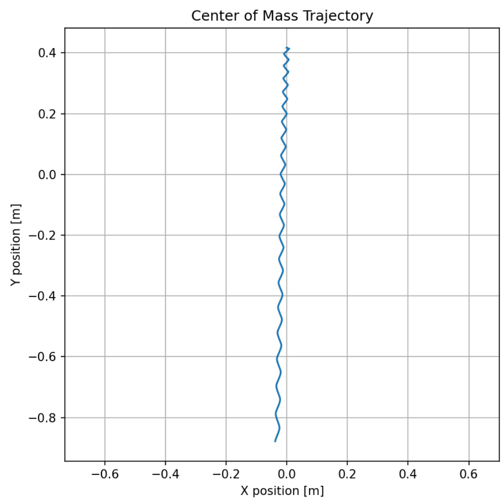


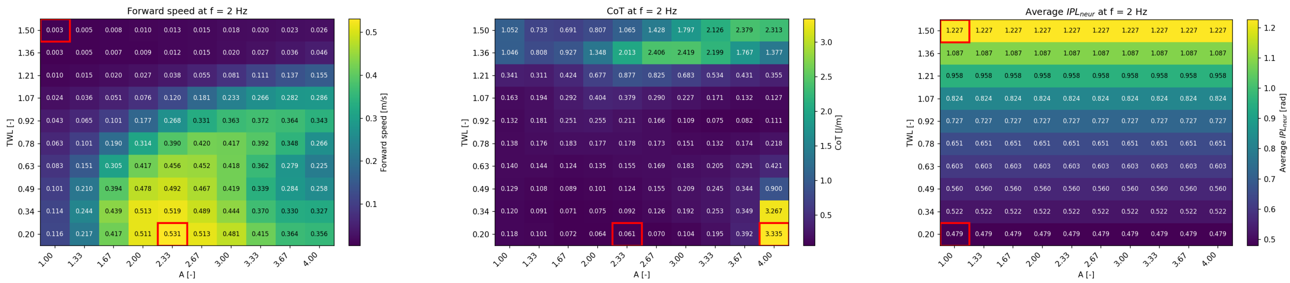
Figure 2: Center of Mass trajectory in 2D

The implemented metrics (NM2, MM2, and MM4) seem to correctly evaluate the performance of the wave controller. The evolution of the joints angles (Fig.1) show clear periodic oscillations with similar frequencies and amplitudes across joints, and a consistent phase lag between neighbouring joints, which matches the expected travelling wave pattern. In addition, the CoM trajectory illustrate on Fig.2 is almost a straight line with very little lateral deviation, which indicates efficient forward motion and good symmetry. Overall, the results are consistent with what we expect from a swimming behaviour, so the metrics appear to be correctly implemented.

Question 1.3 : Influence of wave controller parameters on robot swimming performance

The grid search were done on 10 point for each range ($A \in [1, 0, 4.0]$ and $TWL \in [0.2, 1.5]$) and the results are shown in Fig. 3.

Fig. 3a shows that lower TWL increases forward speed. The optimal speed (0.531 m/s) occurs at a low TWL (0.20) and medium amplitude (A = 2.33). Similarly, CoT (Fig. 3b) generally decreases at lower TWL and medium amplitude, reaching a minimum of 0.118 J/m. Lastly, average IPL$_{neur}$ (Fig. 3c) correlates positively with TWL but remains independent of amplitude.



(a) Forward speed grid search (b) CoT grid search (c) IPL$_{neur}$ grid search

Figure 3: Grid search for question 1.3

2 CPG Controller

Question 2.1 : CPG controller implementation

In this first part, we implement a CPG controller for the Polymander. The body consists of a dual chain of 16 oscillators along the body and 16 oscillators along the four limbs. For this first project, we assume that the limb network is saturated with zero amplitude, and we therefore only consider the axial network.

We first implemented the ODE and the derivative updates. The following graphs illustrate the evolution over time of the oscillator states (θ and r), the sum and difference of the muscle outputs, the body joint angles, and the robot's center of mass (CoM) trajectory over 5 seconds of simulation.

Oscillators states

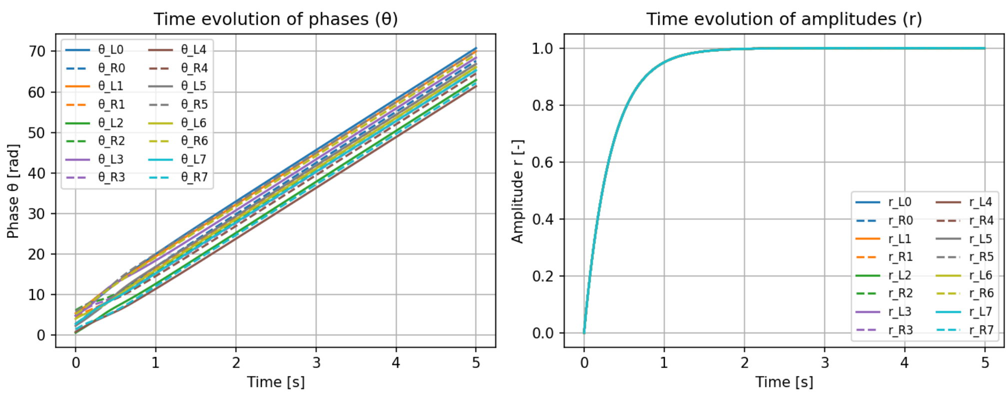


Figure 4: Oscillators phase θ and amplitudes r over time

From Fig. 4, we observe that all the oscillators' phases evolve linearly over time (modulo 2π). The solid lines represent the oscillator phases on the left side of the body, while the dashed lines correspond to the oscillator phases on the right side. The phase increases continuously at a constant rate determined by the frequency, so once the system reaches a steady state, θ grows linearly. This results in straight lines when plotted over time, which indicates stable and regular oscillations.

Also in Fig. 4, we can observe that all the oscillators' amplitudes converge to 1 over time. Indeed, with CPG the amplitude r follows a differential equation that pushes it toward a predefined target value. After a short transient phase, all oscillators settle at this value, which shows that the system has reached a steady and stable oscillation.

Muscles outputs

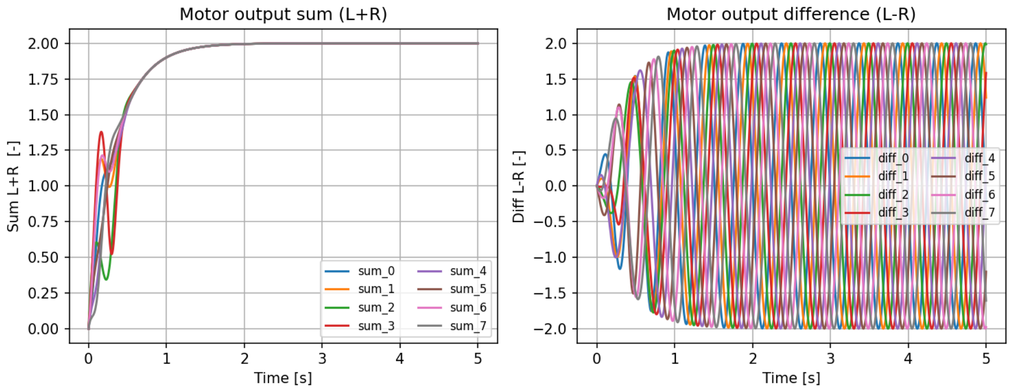


Figure 5: Difference and sum of the muscles outputs

The sum of the muscle outputs in Fig. 5 (left) remains close to two over time. This is expected because the muscle activation function includes a constant offset, which introduces a positive baseline in both left and right outputs. As a result, even though the oscillatory components are out of phase, they do not fully cancel out. The difference of muscle outputs Fig. 5 (right) represents the effective driving signal for motion generation. It illustrates a clear oscillatory pattern, typically close to a sinusoid, since the left and right muscles are activated in opposition. This alternating activity is what produces the rhythmic bending of the body.

Overall, these two signals confirm that the CPG is correctly generating coordinated and symmetric motor patterns.

Body joint angles

The body joint angles Fig.6 exhibit regular oscillatory behaviour driven by the CPG. The active joints show clear and smooth sinusoidal motions, consistent with the rhythmic motor commands generated by the oscillators.
In contrast, joints 5 and 9 are passive joints and do not oscillate in the same plane as the active ones. As a result, their motion is not directly aligned with the main actuation direction, which leads to significantly weaker oscillations.

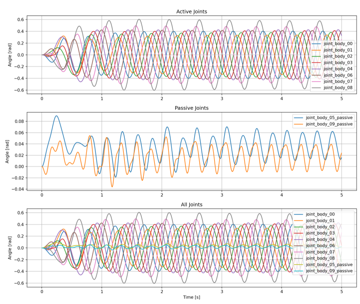


Figure 6: Body joint angles

Center of Mass

The Center of Mass (Fig.7) trajectory in 2D shows a smooth and continuous motion over time, resulting from the coordinated oscillations generated by the CPG. The periodic body deformations translates into a forward displacement of the robot.

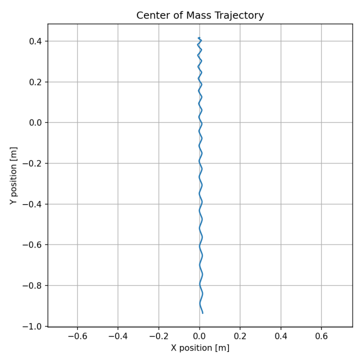
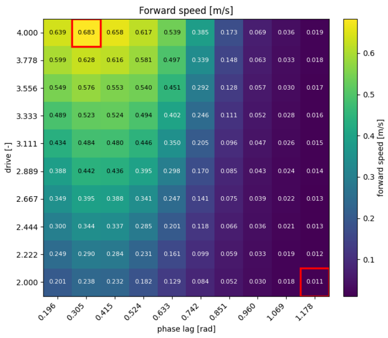


Figure 7: Representation of robot's center of mass in 2D

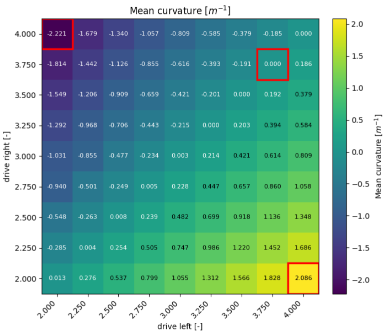
Question 2.2 : Drive values and phase bias effects

The grid search for Fig. 8a and 8b, where done with 10 points for each range ($d = d_{left} = d_{right} \in [2.0, 4.0]$ and $PL \in \left[\frac{\pi}{2n_{joints}}, \frac{3\pi}{n_{joints}}\right]$), while the ones for figures 8c and 9 where done with 9 points for each range($d_{left}, d_{right} \in [2.0, 4.0]$).



(a) Forward speed grid search

(b) CoT grid search



(c) Mean curvature grid search

Figure 8: Grid search for question 2.2

Forward speed correlates positively with drive (d) and negatively with phase lag (PL), as shown in Figure 8a. The maximum speed occurs at d = 4.000 and PL = 0.0305, while the minimum is at d = 2.000 and PL = 1.178. CoT remains generally low (Fig. 8b), except for a sharp increase at high drive and low PL (which also yields maximum speed). The optimal efficiency (lowest CoT) occurs at d = 3.778 and PL = 0.633, while still maintaining a high forward speed (0.497 m/s).

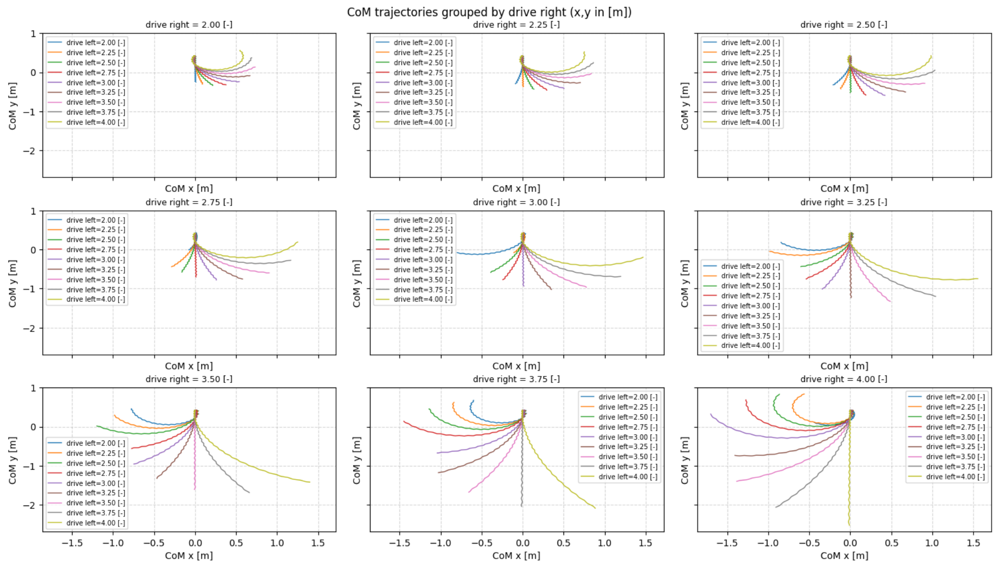


Figure 9: CoM trajectory for drive grid search

Fig. 8c and 9 demonstrate that equal lateral drives yield zero curvature (a straight trajectory). Increasing the difference between left and right drives increases absolute curvature, producing tighter turns. Furthermore, the robot systematically steers toward the side with the higher drive.

Question 2.3 : animal locomotion performance vs simulation

The body joint angles of the simulation and the animal show similar periodic oscillations (Fig.10), with a difference in amplitude but a similar frequency as it can be see in (Table. 1).

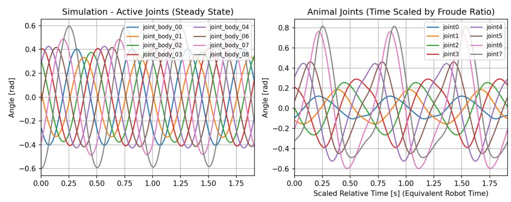


Figure 10: Comparison of body joint angles between real animal and our simulation

	Frequency [Hz]	Amplitude [rad]	IPL [rad]
Simulation	2.00	0.40	5.51
Animal	1.63	0.37	5.36
Error	**0.38**	**0.13**	**0.33**

Table 1: Frequency, amplitude and IPL comparison between animal and simulation

The quantitative comparison presented in Table 1 shows that the implemented controller successfully reproduces the general structure of animal locomotion and most of the root mean square error remains relatively low. This could be improved by tuning the simulation with different values, to better mimic the animal swimming.

Overall, the comparison of body joint angles indicates that the implemented controller captures the main locomotion pattern of the animal, but with some differences in frequency, amplitude, and phase coordination, leading to an imperfect but qualitatively consistent reproduction of the gait.

3 Sensory Feedback

Question 3.1 : Stretch feedback implementation and influence on locomotion performance

In this section, we extend the CPG controller by incorporating ipsilateral stretch feedback. We compare two configurations: open-loop ($w_{ipsi} = 0$) and closed-loop ($w_{ipsi} = 3.0$).

Oscillators states

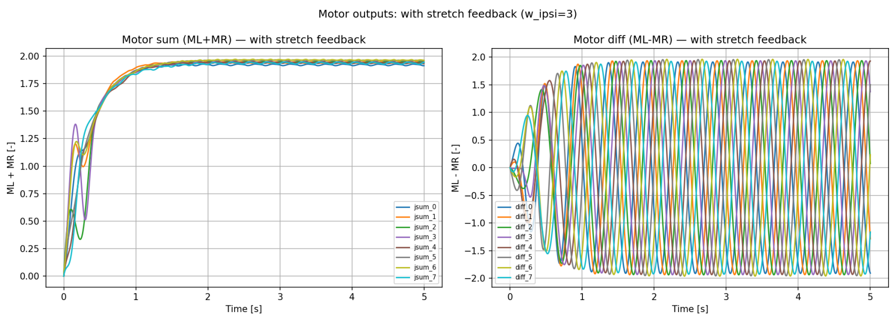


Figure 11: Oscillators phase θ and amplitudes r over time, with stretch feedback

We compare oscillator states with (Fig.11) and without (Fig.4) sensory feedback. The phase θ shows no visible difference when adding feedback, although in theory, sensory feedback can accelerate or decelerate the CPG phase while keeping it synchronized with body movements [1].
However, the amplitude r shows an interesting change of behaviour in steady state. Instead of smoothly converging to its nominal value, like in Fig.4, the closed-loop version shows sinusoidal fluctuations settling around a smaller value than the open-loop nominal value. This modulation fits what we expect from sensory feedback, as stretch-sensitive cells in lampreys and zebrafish [1] are known to modulate the gain of muscle contractions locally. In our model, this is explained by the feedback term $s_i cos(\theta_i)$ continuously modulating the amplitude instead of letting it converge.

Body joints angles

Comparing joint angles with (full lines) and without (dashed lines) feedback, for a given joint, the two curves are indistinguishable during the transient phase (Fig.12). When approaching steady state, the

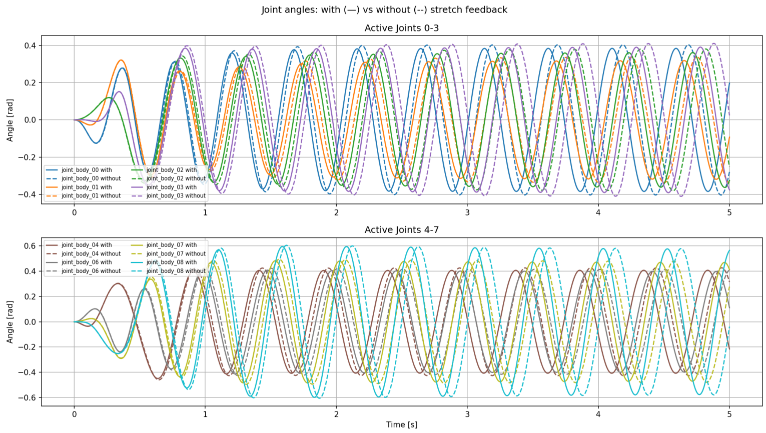


Figure 12: Body joint angles, with and without stretch feedback

curves of the closed-loop system start shifting ahead of the open-loop ones more and more over time, which indicates an accumulating phase difference caused by a frequency difference between the two conditions. Therefore, even if it was too small to be seen in Fig. 11, there is indeed a modulation of the frequency made by stretch feedback.
A difference in amplitudes is visible, with the closed-loop system showing overall smaller amplitudes than the open-loop system, which is consistent with the amplitude modulation discussed above.

Muscles outputs

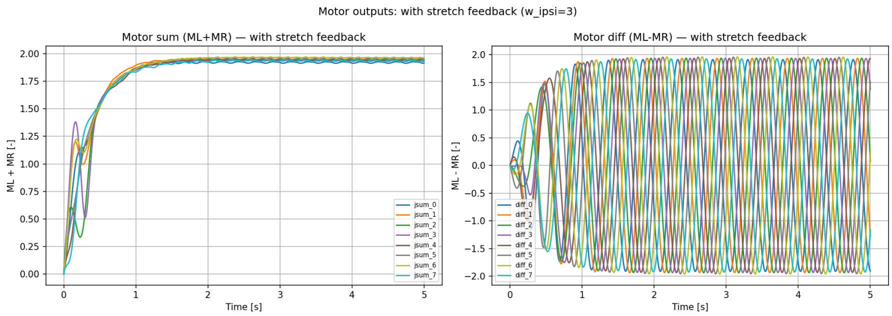


Figure 13: Difference and sum of the muscles outputs, with stretch feedback

Fig.13 confirms again the phase and amplitude modulation by the stretch feedback. The sum of motor outputs (on the right) converges around a slightly smaller value than the open-loop version (Fig. 5), with small oscillations. The motor output difference (on the left) shows slightly smaller amplitude with feedback, not quite reaching the peak values of the open-loop version (Fig. 5).

Center of Mass

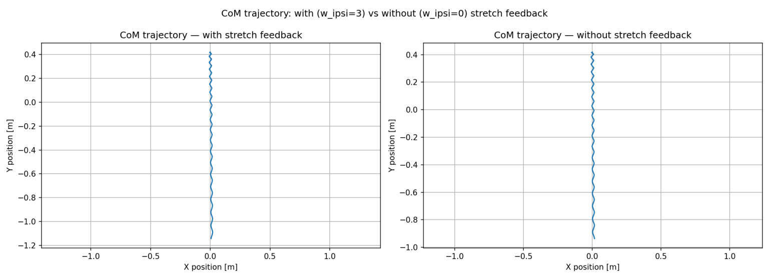


Figure 14: Representation of robot's center of mass in 2D, with and without stretch feedback

In Fig. 14 we notice that, while the trajectory's overall shape stays the same, the closed-loop version reaches a greater length, of more than 1 meter, than the open-loop version that does not even reach 1 meter in the same number of simulation steps (*runtime_n_iterations*=2501).

Metric values

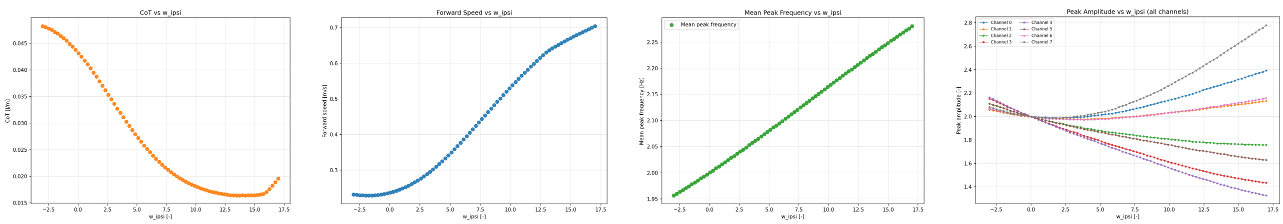
Metric	**No feedback** ($w^{\text{ipsi}} = 0$)	**With feedback** ($w^{\text{ipsi}} = 3.0$)
Neural frequency f_{neur} [Hz]	2.000	2.047
Neural amplitude A_{neur} [-]	2.000	1.939
Forward speed v_{fwd} [m/s]	0.226	0.272
Cost of transport CoT [J/m]	4509.820	3604.291

Table 2: CPG controller performance with and without stretch feedback ($w^{\text{ipsi}} = 3.0$ vs $w^{\text{ipsi}} = 0$).

The following metrics were computed over 20'001 steps, skipping the transient phase, allowing for a perfect 2.000 Hz and 2.000 [-] for neural frequency and amplitude respectively, when using only CPG (Table 2). Once we introduce sensory feedback, the neural frequency increases to 2.047 Hz, explaining the constant phase shift observed in Fig. 12. The neural amplitude decreases to 1.939 [-], accounting for the overall lower amplitudes we noticed for the closed-loop version in Fig. 11, 12, 13.
As expected from Fig. 14, the forward speed increases from 0.226 [m/s] to 0.272 [m/s] when adding feedback. Finally, cost of transport decreases from 4509.820 [J/m] to 3604.291 [J/m] when using stretch feedback. These two last metrics strongly support the idea that combining of CPG and sensory feedback can produce more efficient locomotion.

Question 3.2 : Ipsilateral feedback tuning influence

The following results highlight how increasing sensory feedback strength (w_{ipsi}) influences locomotion performance and the underlying neural dynamics.



(a) CoT (b) Forward speed (c) Peak frequency (d) Peak amplitude

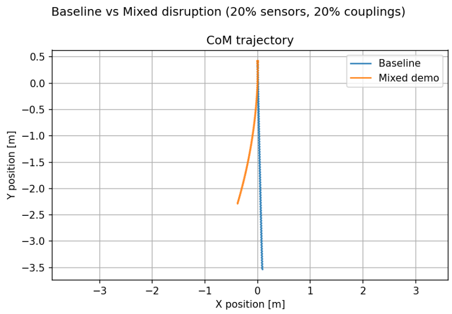
Figure 15: Grid search for question 3.2

The cost of transport (Fig.15a) and forward speed (Fig.15b) grid search mirror each other. This fits with the idea that both a lower cost of transport and a higher forward speed translate for a more efficient locomotion, which is what we observe overall when we increase the value of w_{ipsi}, and therefore the strength of the sensory feedback. The slopes are the most steep for $w_{ipsi} \in [0.0, 12.5]$, the ends are overall flatter or flat. Cost of transport even increases back after $w_{ipsi} = 15$, which suggests that too much sensory feedback can become counter productive. This makes the results for a $w_{ipsi} \cong 15$ the best in terms of efficiency.
Neural frequency increases linearly with w_{ipsi} (Fig.15c), probably because imposing faster bending of the body forces the CPG to speed up through sensory feedback.
When $w_{ipsi} = 0$, all channels meet at a peak amplitude of 2.0 (Figure 15d), as found earlier in Table 2 for the open-loop system. The channels diverge as w_{ipsi} increases or decreases, which tells us that this modulation is position dependant. The channels represent joint pairs along the salamander body, their numbers indicate their positions on the robot. The channels increasing above 2.0 are the ones close to the head (0,1) or to the tail (6,7), while the ones decreasing below 2.0 are the pairs closer to the center

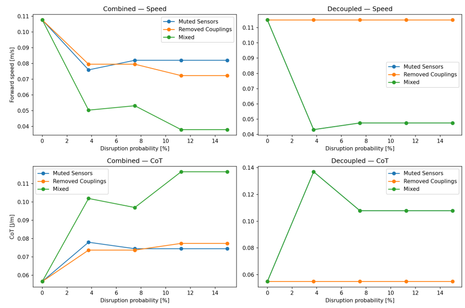
of the body (2,3,4,5). The centered body joints experience the largest bending during locomotion, considering they are at the antinode of the travelling wave. Because they have the largest bending, they receive the strongest stretch feedback, which actively reduces their amplitude. Alternatively, the joints on extremities bend less, therefore receive weaker feedback, and their amplitude increases above nominal value.

Question 3.3 : Neural disruptions impact

The following analysis investigates how various disruptions affect locomotion performance, trajectory stability, and efficiency.



(a) Representation of robot's center of mass in 2D, with neural disruptions



(b) Speed and cost of transport comparison of combined and decoupled neural disruptions

Figure 16: CoM trajectories and results from the different combination of disruptions

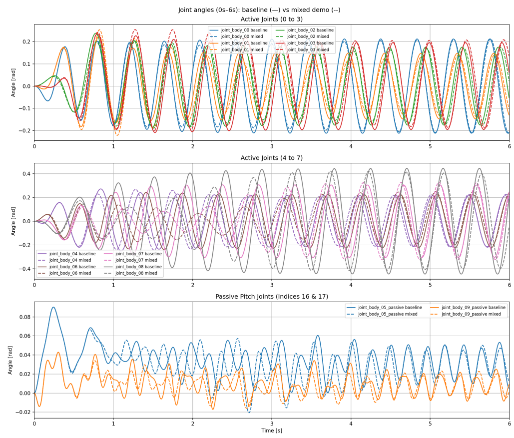


Figure 17: Body joint angles, with neural disruptions

Fig.16a shows that with mixed disruptions, the trajectory drifts sideways and does not go as far as the baseline, meaning these disruptions reduce forward speed and compromise straight line locomotion. Joint angles remain qualitatively similar but show amplitude and period variations caused by the disruptions (Fig. 17). The combined cases in Fig. 16b show the effect of both disruptions on sensors and couplings, separately and combined. For both forward speed and cost of transport, the disruptions each affect the metrics partially, and then create a bigger impact when combined. In the decoupled case, the removed couplings is flat considering the ipsilateral couplings are already absent. Therefore the *Muted Sensors* and *Mixed* curves overlap and reveal a lower forward speed and higher cost of transport. Overall, disruptions degrade the performance of the salamander robot and show less efficient results.

Generative AI Declaration

For this project, we used Generative AI for two main tasks. First, it helped us better understand how certain variables evolve, such as joint angles or oscillator phases and amplitudes. Second, it was used to generate plots and organize them into subplots. In particular, it enabled an easy separation of the robot's left and right components, which significantly improved readability and saved time in understanding the simulation's behavior.

References

[1] EPFL. Reflexes and sensory feedback. https://moodle.epfl.ch/pluginfile.php/3530179/mod_resource/content/1/lecture8_Reflexes_Sensory_Feedback.pdf, 2023.